

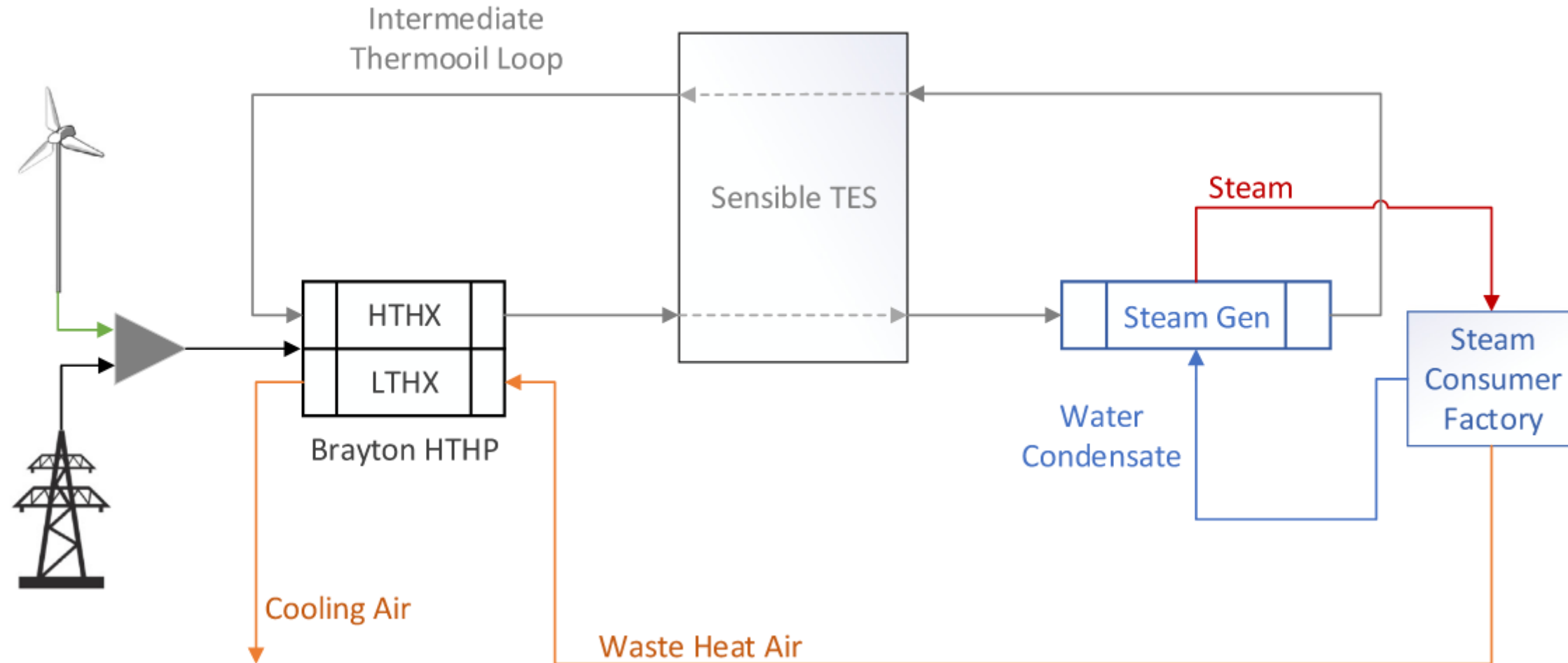
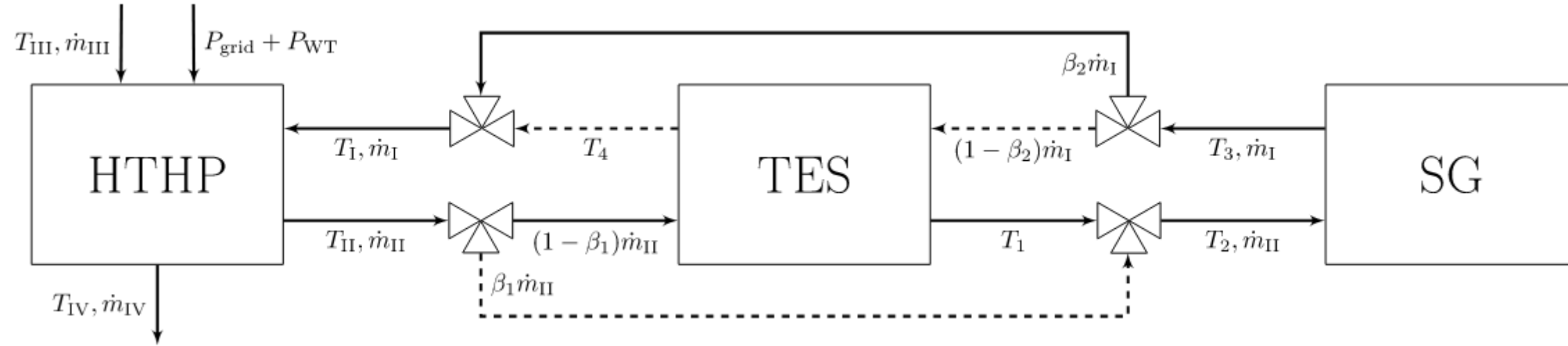


Illustration of the investigated industrial energy conversion system for electrified steam generation. The system consists of a HTHP, a TES and a SG, where the HTHP is powered by electricity from a wind turbine or the power grid. The HTHP uses waste heat air stream as a heat source and enables charging and discharging of the TES via an intermediate thermal oil stream. Furthermore, constant heat demand for the steam consumer factory must be satisfied.



Detailed flow chart of studied system including HTHP, TES and SG. HTHP and SG are represented by polynomial surrogate models, while TES is modeled by an effectiveness model. An adjustable fluid bypass $\beta_1 \in [0, 1]$ is used to control the heat input to the TES and SG, conversely the bypass $\beta_2 \in [0, 1]$ regulates the heat input to the HTHP depending on the thermal state T_s of the TES and SG outlet stream. The solid and dashed lines indicate the charging and discharging mode in a simplified way. In addition, simultaneous charging and discharging is not allowed. For example, charge mode is $\beta_1 \in [0, 1)$, $\beta_2 = 1$, discharge mode implies $\beta_2 \in [0, 1)$, $\beta_1 = 1$, and idle mode represents $\beta_1 = \beta_2 = 1$

Minimize **total operational cost**:

➤ **Cost depends on:**

- Electricity purchased from the grid P_{grid}^k
- Time-varying electricity prices g_{grid}^k

$$\min J(P_{\text{grid}}) = \sum_{k=1}^n P_{\text{grid}}^k g_{\text{grid}}^k \Delta t$$

Constraint-1

Power Balance Constraint

➤ Ensures:

- Energy supply = energy demand
- Total power input (grid + wind) must equal to:
 - Power required by the heat pump system

➤ Includes:

- Grid power
- Wind turbine power
- Heat pump consumption

$$P_{\text{grid}}^k + P_{\text{WT}}^k = 3F_{\text{HTHP}}(T_I^k, \dot{m}_I^k, T_{III}^k, R^k)$$

Constraint-2,3

- These are **nonlinear surrogate models**
- They compute temperatures based on:
 - Flow rate
 - Temperatures
 - Compressor speed

$$T_{II}^k = F_{\text{HTHX}}(T_I^k, \dot{m}_I^k, T_{III}^k, R^k)$$

$$T_{IV}^k = F_{\text{LTHX}}(T_I^k, \dot{m}_I^k, T_{III}^k, R^k)$$

Constraint-4,5

- Control variables $\beta_1, \beta_2 \in [0,1]$
- Define **how heat is routed** in the system

$$T_2^k = T_{II}^k \beta_1^k + T_1^k (1 - \beta_1^k)$$

➤ Control whether energy:

- Goes to storage
- Bypasses storage
- Returns to the heat pump

$$T_I^k = T_3^k \beta_2^k + T_4^k (1 - \beta_2^k)$$

Constraint-6,7

- Relate **temperature and mass flow rate**
- Ensure required **steam temperature conditions** are satisfied.

$$T_2^k = 201.92 + \frac{1819.32}{3\dot{m}_{II}^k}$$

$$T_3^k = 196.3 - \frac{188.4}{3\dot{m}_I^k}$$

Constraint-8,9

➤ Thermal Energy Storage (TES)

- Charging
- Discharging

➤ Define heat flow into and out of storage:

- TES stores excess heat and releases it when needed.

$$\dot{Q}_{s,\text{ch}}^k = 3\dot{m}_{II}^k c_{p,f} (T_{II}^k - T_1^k) (1 - \beta_1^k)$$

$$\dot{Q}_{s,\text{dch}}^k = 3\dot{m}_I^k c_{p,f} (T_4^k - T_3^k) (1 - \beta_2^k)$$

Constraint-10

➤ No Simultaneous Charge & Discharge

- Prevents charging and discharging at the same time

$$\dot{Q}_{s,\text{ch}}^k \dot{Q}_{s,\text{dch}}^k \leq \gamma$$

Constraint-11,12,13

➤ Storage Temperature Dynamics

- Storage temperature evolves over time(dynamic behaviour)

$$T_1^k = T_{II}^k - \epsilon_{\text{ch}} (T_{II}^k - T_s^{k-1})$$

$$T_4^k = T_3^k - \epsilon_{\text{dch}} (T_3^k - T_s^{k-1})$$

$$T_s^k = T_s^{k-1} + \frac{\dot{Q}_{s,\text{ch}}^k - \dot{Q}_{s,\text{dch}}^k}{m_s c_{p,s}} \Delta t$$

Constraint-14

➤ Boundary Conditions

- Initial and final storage temperature must match
- Ensures **cyclic daily operation**

$$T_s^0 = \tilde{T}^0, \quad T_s^n = T_s^0$$

Constraint-15

➤ Operational Limits

➤ Limits on:

- Temperature
- Flow rate
- Compressor speed

$$T_I^k \in [177, 250], \quad \dot{m}_I^k \in [5, 16]$$

$$T_{III}^k \in [60, 100], \quad R^k \in [0.8, 1.53]$$

Constraint-16

➤ Time Discretization

- System is optimized over discrete time steps

$$t_k = \Delta t \, k, \quad k = 1, \dots, n$$

Thank You!

I appreciate your time and attention.